

TELEMETRY DATASET ANALYSIS

Sleep Debt & Screentime Physiology

Comprehensive review of device habits, app arousal, and sleep architecture degradation.

Expanded Dataset Scope

8.5_k

PARTICIPANT PROFILES

24

MONITORED FEATURES

14

DAYS TRACKED

98%

LOG COMPLETION

Cohort Baselines

Demographic and chronotype distribution across the validated sample.

Median Age

28.4 Years

Gender Skew

54% Female

Primary Chronotype

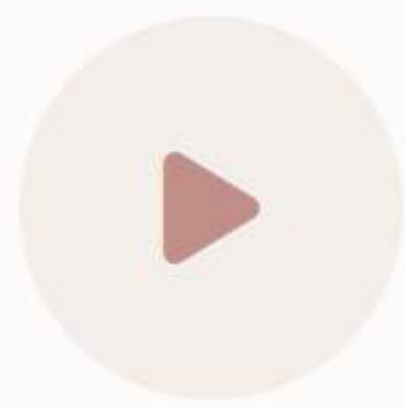
Night Owl (61%)

The Pre-Sleep Window

48_m

Average active screentime while physically in bed.

Cognitive Arousal by App Class



Short-Form Video

Maximum neuro-arousal via rapid context switching.



Social Feeds

Infinite scroll mechanics delaying sleep initiation.



E-Readers

Lowest arousal. Minimal latency disruption.

Device Settings

- **Blue Light Filters**
Reduces sleep latency by 28 minutes on average.
- **Max Brightness**
Suppresses natural melatonin onset by 1.5 hours.
- **Dark Mode UI**
Negligible physiological benefit without dimming.





Deep Sleep Collapse

High-brightness screen exposure post-10 PM severely degrades sleep architecture.

-22%

Reduction in Deep Sleep Phase.

+1.4

Additional waking events per night.

Sleep Latency Impact

High Brightness + No Filter

84 mins



Low Brightness + No Filter

52 mins



Dimmed + Blue Light Filter Active

20 mins



Next-Day Morning Consequences

Mild Sleep Deficit

Bedtime Phone < 30 mins

Snooze Count 1 time

Grogginess Score Low

Severe Sleep Debt

Bedtime Phone 120+ mins

Snooze Count 4+ times

Grogginess Score Severe

The Caffeine Compensation Loop

310_{mg}

Average next-day intake for the Severe Debt cohort.

The "Weekend Recovery" Myth



Sleep Bingeing

Does not reverse cumulative cognitive deficits.



Circadian Shift

Late weekend nights permanently delay melatonin phase.



Sunday Insomnia

Spikes sharply due to weekend phase misalignment.



Predictive Mitigation

Modeling behavioral interventions and hardware automation for
sleep recovery.

Image Sources



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